

Adv C Module

1. Subjective

2.1 Basic Refreshers

1. What are type qualifiers?
2. What are bitwise operators? Explain with examples.
3. Explain bit fields and their use cases.
4. What are the data types?
5. What is the output of the below code

```
while (j <= n) {  
    j = j * 2;  
}
```
6.

```
char str1[] = "ARMV8";  
char str2[] = "RISCV";  
char *sp1 = str1, *sp2 = str2;  
while (*sp1++ = *sp2++);  
printf("%s\n", str1);
```
7. Consider two binary operators, $\uparrow$ and $\downarrow$, with the following properties:
 - Operator $\uparrow$ has higher precedence than $\downarrow$
 - Operator $\uparrow$ is right associative
 - Operator ' $\downarrow$ ' is left associative

Evaluate the expression:

$7 \downarrow 3 \uparrow 5 \downarrow 3 \downarrow 2$

8.

```
int func(void)  
{  
    static int x = 1;  
    x++;  
    return x;  
}  
int main()  
{  
    int a, b;  
    a = func() + func();  
    b = func() + a;  
    printf("%d", b);  
}
```

 What is the output of the above code?

2.2 1D Pointers and Functions

1. What is a pointer? Explain types of pointers.
2. `#include <stdio.h>`

```
int main() {  
    char *c[]={"HELLO", "ENTER", "POINT", "FIND"};  
    char **cp[] = {c+3, c+2, c+1, c};  
    char ***cpp = cp;  
    printf("%s\n", **++cpp);  
    printf("%s\n",*--*++cpp+3);  
    printf("%s\n", *cp[-1][-1]++);  
    return 0;  
}
```

2.3 Storage classes and memory segments

1. Explain memory segments.
2. What is the BSS segment, and what is its purpose?
3. Explain storage classes in detail.
4. What is the difference between local static and global static variables?
5. What is the size of the stack? Who decides it?

2.5 2D Pointers and DMA

1. What is the difference between an array of pointers and Pointer to an array?
2. What is Dynamic memory allocation with a real-time example?
3. Explain the function pointers.
4. What is the difference between an array and a structure?

2.6 Preprocessing

1. What are the stages of compilation?
2. What is the difference between `#include <stdio.h>` and `#include "types.h"`?
3. Where are `printf()` and `scanf()` stored?

2.7 UDT

1. What is the difference between structure and union?
2. What is structure padding? Why is it required?
3. What is endianness? How do you detect endianness using a union?
4. What is a structure pointer example?
5. What are macros?
6. What is the difference between an Inline function and Macro.

2.8 Miscellaneous

1. Explain volatile. Why do we use volatile? Where is it mandatory?
2. Explain the conditional compilation and its use cases.
3. Explain the volatile and const keywords.

2.9 FILE I/O

1. Explain file handling concepts.

2.10 projects

1. Explain the steganography project.
2. Explain the MP3 projects.
3. In a Steganography project, restrict secret files to only 3 lines.
4. Why is data encoded in the LSB instead of the MSB in image steganography?

2. Programming

1. WAP to swap two numbers without using an extra variable.
2. WAP to set, get, clear, and toggle the nth bit.
3. WAP to set all odd bits without hard-coding values like 0xAA.
4. WAP to swap odd and even bits in a 32-bit number.
5. WAP to reverse an array.
6. WAP to reverse the word
Input: Welcome to Alif
Output: Alif to Welcome
7. WAP to check whether a string is a palindrome.
8. Write a program to reverse a string.
9. Write a program to reverse a string using pointers.
10. Write a program to check whether a number is odd or even.

11. WAP to swap two numbers without using an extra variable.
12. WAP to find the factorial using the ternary operator.
13. WAP to print "Hello World" without using a semicolon.
14. WAP to find the second largest number in an array.
15. WAP to print a number in binary format.
16. WAP to swap nibbles of a byte.
17. WAP to determine whether the system is little endian or big endian.
18. WAP to calculate factorial.
19. WAP to generate the Fibonacci series.
20. WAP to find the sum of two matrices.
21. WAP to determine whether the system is little-endian or big-endian.
22. You are given an array a of n elements. In one operation, you can choose an integer $x = a[i]$ and replace all elements equal to x with 0. Return the minimum number of operations required to make the array ascending.
Example:
Input: [3, 3, 2]
Output: 1
Explanation: If $x = 3$, all 3s become 0 $\rightarrow$ [0, 0, 2]. Write a program to solve this problem.
23. Two players, Alex and John, play a game starting with an integer n . Players alternate turns. On each turn, a player can: Divide n by any odd divisor greater than 1 (including n if n is odd), OR Subtract 1 from n , as long as $n > 1$. The game ends when $n = 1$. The player who cannot make a move loses. Write a program to determine the winner.

MC Module

1. Basics

1. What is the difference between Microcontroller and Microprocessor.
2. What is the difference between Harvard and Von Neumann architecture?
3. How do you configure GPIO pins as input/output?
4. If all ports are occupied, how will you connect a new peripheral?
5. How many GPIO pins are available in your controller?
6. What is cache memory? Types of memory. Why is a cache required?
7. Explain MOSFET, BJT, IGBT, and Thyristor.
8. What is the difference between MOSFET and IGBT?
9. Explain inverter and diode.
10. What is an FPGA?
11. What are Decoupling Capacitors?
12. Explain LED interfacing with a microcontroller and draw the circuit diagram.

2. Interrupts

1. What is an interrupt?
2. How is an interrupt initiated?
3. What is the difference between polling and interrupt?
4. Explain the Back-to-back interrupts and Nested interrupts
5. Explain the reset interrupt and interrupt priority.
6. What are timers and counters?
7. What is the difference between a timer and a counter?
8. Explain EEPROM configuration.

3. Projects

1. Explain the CAN-based automotive dashboard project.
2. Explain the car black box project.

4. ADC

1. What is ADC and why is it needed?
2. What types of signals does an ADC convert?
3. Explain the working principle of an ADC.
4. What is ADC resolution?
5. What is meant by 8-bit, 10-bit, and 12-bit ADC?

5. Embedded Systems

1. What is an embedded system? Give an example.

6. Protocols

1. Explain the UART protocol with a frame format.
2. Explain the I2C protocol with a frame format.
3. Explain the SPI protocol with a frame format.
4. Explain the CAN protocol with a frame format.

7. PWM

1. What is PWM and its applications?
2. What happens when the duty cycle is increased?
3. What happens if PWM frequency is too high or too low?
4. Which peripheral is used to generate PWM?
5. What registers are used for PWM configuration?

8. Programming

1. WAP to generate an interrupt using a timer.
2. Write an LED toggle program.

CPP Module

1. Explain the difference between Structure and Class.
2. What challenges did you face during your C++ project implementation?
3. What is the difference between C and C++?
4. What is the difference between function overloading and Function overriding?
5. What is multithreading in C++?
6. WAP to reverse an array using C++.
7. Explain polymorphism in C++ with an example.
8. What are a constructor and a destructor?
9. How do you allocate and deallocate memory in C++?
10. Explain OOPS concepts:
 - a. Encapsulation
 - b. Inheritance
 - c. Polymorphism
 - d. Abstraction

11. What are virtual functions? Explain with an example.
12. Add $1 + 3f$ and $2 + 4f$ without using operator overloading.
13. Explain STL concepts:
 - a. Vector
 - b. Map
 - c. Iterator

DS Module

1. Basics

1. Explain data structures. What data structures have you worked with?

2. Linked lists

1. Explain Singly Linked List (SLL).
2. Explain arrays and linked lists with time complexity.
3. How do you analyse and manage time complexity?
4. WAP to insert a node at the last position.
5. WAP to insert a node in the middle.
6. Declare a linked list structure and find its size.
7. Design a structure Node to represent a tree-like data structure where:
 - Each node stores integer data
 - Number of branches is decided at runtime
 - Minimum branches = 1, Maximum branches = 5
8. Write a program using a doubly linked list:
 - a. Insert before
 - b. Insert after
 - c. Delete a node
 - d. Display list

3. Stack

1. Explain stacks and queues with real-time examples.
2. Explain the LIFO principle.
3. What are the basic stack operations?
4. What are stack overflow and stack underflow?
5. How do you implement a stack using an array?
6. How do you implement a stack using a linked list?

4. Searching and Sorting Techniques

1. Explain searching and sorting algorithms.
2. Explain linear search.
3. Time complexity of linear search (best, worst, average).
4. What is the prerequisite for binary search?
5. Time complexity of binary search.
6. Can binary search be applied to linked lists? Why?

5. Queue

1. What is a queue?
2. Explain the FIFO principle.
3. What are enqueue and dequeue operations?
4. What is a priority queue?
5. What is a deque (double-ended queue)?

6. Hashing

1. What are hashing, hash functions and hash tables?
2. What makes a good hash function?
3. Advantages and disadvantages of hashing.

7. Project

1. In an APC project, throw an error if operands exceed 5 digits.
2. Explain the inverted search project.

8. Trees

1. Explain trees and their applications.
2. Define root, parent, child, leaf, and height of a tree.
3. What is the difference between a tree and a graph?

LI Module

1. Basics

1. What is RTOS?
2. What is the difference between Hard RTOS and Soft RTOS?
3. What is the output of the code below?

```
int main() {  
    fork();  
    fork();  
    printf("mimyk");  
}
```

4. Explain the Linux root directory structure.

2. System call

1. Explain system calls with a diagram.
2. Why do we need system calls?
3. How does a system call work internally?
4. What is the difference between a system call and a library function?
5. Explain the exit() system call.
6. Explain open(), read(), write(), and close().

3. Networking

1. What is computer networking?
2. What is HTTP?
3. What is FTP?
4. What is DNS?
5. What are select() and poll()?
6. What is the difference between select() and epoll()?
7. What is a three-way handshake?

4. Process

1. What is a process?
2. Explain the process state diagram.
3. Write a program to create a process using fork().
4. What is a zombie process?
5. Write a command to list the processes running during program execution.

5. IPC

1. Explain the IPC mechanisms.
2. What is the difference between pipe and FIFO?
3. What is the difference between a message queue and shared memory?
4. When do you prefer shared memory over message queue?

6. Signal

1. How do you send and handle signals?
2. Explain signalling.

7. Socket

1. Write TCP and UDP server-client programs.
2. Modify TCP/UDP programs to accept command-line arguments instead of hardcoded ports.

8. Project

1. Convert a TFTP project from UDP to TCP.
2. Explain the minishell project.

9. Threads

1. What is a thread?
2. What is the difference between a process and a thread?
3. What are a mutex and a semaphore?
4. What is the difference between a mutex and a semaphore?
5. How do threads behave when a mutex is used?

10. Process and Memory Management

1. What is memory management?
2. What are the different types of memory?
3. Explain virtual memory.
4. What is paging?
5. What is segmentation?
6. What is the difference between static and dynamic memory allocation?
7. What are `brk()` and `mmap()`?

Qt Module

1. What is multithreading, and how do you create multithreading in QT/C++?
2. Write a C++ program to increase the brightness of an RGB image stored in a 3-dimensional matrix.
 - Brightness should be taken from a QML-based front-end widget
 - The output image should be shown in the same widget

General Questions Aptitude

1. How do you create multiple directories using a single command?
2. What is the difference between a processor and a controller?
3. Is your laptop a processor or a controller? Why?
4. How do you print a message in a shell script?
5. What are the functions of \$\$, \$# , and \$ in shell scripting?
6. How will you debug a shell script if an error occurs at line 3?
7. Are you comfortable coding in Python?
8. If someone knows your password, it is not secure. What steps will you take to improve security?
9. How would you rate yourself in Python and Selenium?
10. What are dictionaries in Python?
 - a. What are lists and tuples? Differentiate them.
11. What is GitHub, and how do you use it?
12. Explain the following Linux commands: scp, nproc, uname -a, and ifconfig.
13. Which modules have you learnt?
14. What was your course time period?
15. Explain the projects you have done.
16. Explain your academic projects.
17. Any previous internships?
18. What programming languages do you know?

19. What is the name of the battery (18650)?
20. If given a 3.7V battery, how will you get 5V output?
21. What is a buck converter?
22. What is a boost converter?
23. Are you comfortable with Python programming?
24. Explain dictionaries in Python.
25. Explain lists and tuples in Python.
26. Can you write simple Python scripts?
27. What is WiFi?
28. What is Bluetooth?
29. What is an RF signal?
30. Explain these Linux commands:
 - a. scp
 - b. nproc
 - c. uname -a
 - d. ifconfig
31. Find the odd one out: LINUX, UBUNTU, WINDOWS, MACOS
32. A person walks 10 m north, then turns right and walks 10 m, then turns right again and walks 10 m. Where is the person facing now?
33. Find the next number in the series: 2, 6, 12, 20, ?
34. Find the next number in the series: 1, 2, 4, 8, 7, 14, ?
35. What comes next in the series: A, C, F, J, O, ?
36. If a person covers 600 m in 5 minutes, find the speed in km/hr.
37. A can do a work in 10 days, and B can do it in 15 days. How many days will they take together?
38. How do you check the Ubuntu version?
39. How do you find files in Linux?
40. How do you find a USB device connected to a system?
41. If a USB device is not detected, how would you debug the issue?
42. What is the difference between vim editor and the touch command?